

SURESH A

Senior Full Stack Engineer | Elixir & Phoenix | Vue 3 | PostgreSQL

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with **6 years** of experience architecting and shipping **multi-tenant SaaS** and enterprise platforms end to end. Specialist in **Elixir / OTP** and **Phoenix** for concurrent, fault-tolerant backends, paired with modern **Vue 3** frontends. Delivered 6 production products across EdTech, facility management, climate tech, manufacturing, retail and agri-development, owning the path from client requirement and data model to API, UI, payments and release. Known for reliable payment flows, clean bulk-data pipelines and pragmatic system design.

CORE EXPERTISE

Backend	Elixir, OTP (GenServer, Supervisors, Task), Phoenix, Phoenix LiveView, Ecto, Oban, Node.js, Express.js
Frontend	Vue 3 (Composition API), Vite, Pinia, shadcn-vue, Tailwind CSS, TanStack Table, Zod + VeeValidate, Nuxt.js, Vuetify, ApexCharts
Architecture	Multi-tenant SaaS (schema-per-tenant), REST API design, background job orchestration, single sign-on, RBAC, audit trails
Data	PostgreSQL (schema design, indexing, query tuning, JSONB), Firebase Firestore, S3 / DigitalOcean Spaces, IndexedDB
Integrations	BillDesk, Razorpay, WhatsApp Business API, Google Drive API, loan-provider webhooks, SMTP, SMS / OTP
Security	JWT / Guardian, signed payment callbacks, API-key auth, Firebase Auth, secure byte-range media streaming
Delivery	Git, GitHub / GitLab, Linux servers, Elixir releases, Firebase Cloud Functions, Capacitor (Android), Jira, Agile / Scrum

KEY ACHIEVEMENTS

- Architected a **university ERP, LMS and helpdesk suite** covering admission to graduation for an online degree programme.
- Built a **self-healing payment reconciliation** flow that re-checks pending gateway transactions, so paid students are never left unadmitted.
- Designed **schema-per-tenant** SaaS backends so new client organisations onboard by subdomain with isolated data.
- Improved report and dashboard performance by up to **40%** through schema redesign, indexing and precomputed data.

PROFESSIONAL EXPERIENCE

Software Engineer (Full Stack) — Ardhika Software Technologies Pvt. Ltd., Chennai **Oct 2020 – Present**

Own features end to end across 6 client products: requirements with clients, data modelling, Phoenix APIs, Vue frontends, integrations and production releases. 5,000+ active users across products.

Sastra (ChalknBoard) — University ERP, LMS and Helpdesk

Elixir, Phoenix 1.8, LiveView, Oban, PostgreSQL, Vue 3, shadcn-vue, Tailwind CSS, BillDesk, DigitalOcean Spaces

- Built admission-to-graduation workflows: applications, **BillDesk** fee payments with signed-callback verification, automated receipts, GST, education-loan webhooks, and bulk student promotion and graduation.
- Engineered supervised **GenServer** schedulers and **Oban** workers for payment reconciliation, compliance validation and government data uploads, isolated so a failing job never affects the API.
- Delivered a multi-tenant **LMS** with ERP single sign-on, online exams and assignments, secure byte-range video streaming from object storage, and a resumable Google Drive bulk course importer.
- Redesigned course resolution into a precomputed exam-paper table, replacing multi-table joins on every request.

Carbon (Zasti) — Climate-Tech GHG Emission Accounting SaaS

Elixir, Phoenix, Oban, PostgreSQL (schema-per-tenant), Vue.js, Vuetify, Chart.js

- Built a **Scope 1, 2 and 3** emission calculation engine across 10 categories with unit normalisation and versioned emission-factor datasets, so past reports stay reproducible.
- Automated **CDP and TCFD** disclosure reports and peer benchmarking dashboards; added transactional, Oban-scheduled subscription invoicing.

Inconn — Facility Management (CMMS) Platform

Elixir, Phoenix, GenServer, PostgreSQL (schema-per-tenant), Nuxt.js, Vue.js

- Developed preventive-maintenance scheduling that auto-generates time-zone-aware work orders, plus checklists, work-permit and LOTO approvals, and QR-code complaint ticketing.
- Built SLA scorecards (MTBF, MTTR, on-time completion), alert escalation, rosters, attendance and inventory modules, improving workforce coordination by 30%.

RoloCRM — Retail Loyalty and Engagement SaaS

Node.js, Express.js, Firebase Cloud Functions, Firestore, Vue.js, Phoenix LiveView, Razorpay, WhatsApp Business API

- Built a POS-integrated CRM with loyalty points, tiers and a dynamic customer segmentation engine, lifting customer retention by 20%.
- Automated scheduled **WhatsApp** campaigns with delivery tracking, and a LiveView subscription console with Razorpay billing.

Jewelflow — Jewellery Manufacturing ERP

Elixir, Phoenix, PostgreSQL, Vue.js, Vuetify

- Built department-wise production tracking with QR-tagged orders, an 8-state workflow, full audit history and weight-accurate split and merge of spoilt orders, improving operational efficiency by 25%.

Saro (EZCF) — Agri Programme Monitoring Platform

Elixir, Phoenix, PostgreSQL, Vue.js, Capacitor (Android), IndexedDB

- Built an **offline-first** Android app that captured 1,000+ farmer records without connectivity and syncs safely, plus gender- and youth-disaggregated donor impact reports.

Data Entry Operator — Axis Bank (via Sun Facilities Services), Chennai

Mar 2019 – Jul 2020

- Maintained customer and transaction records in the core banking system to audit and compliance standards, the foundation of the data-validation and payment-reconciliation focus in my engineering work.
- Produced transaction-volume reports for branch management and resolved customer account queries.

EDUCATION

Bachelor of Science, Computer Science — Hindustan College of Arts & Science, Chennai